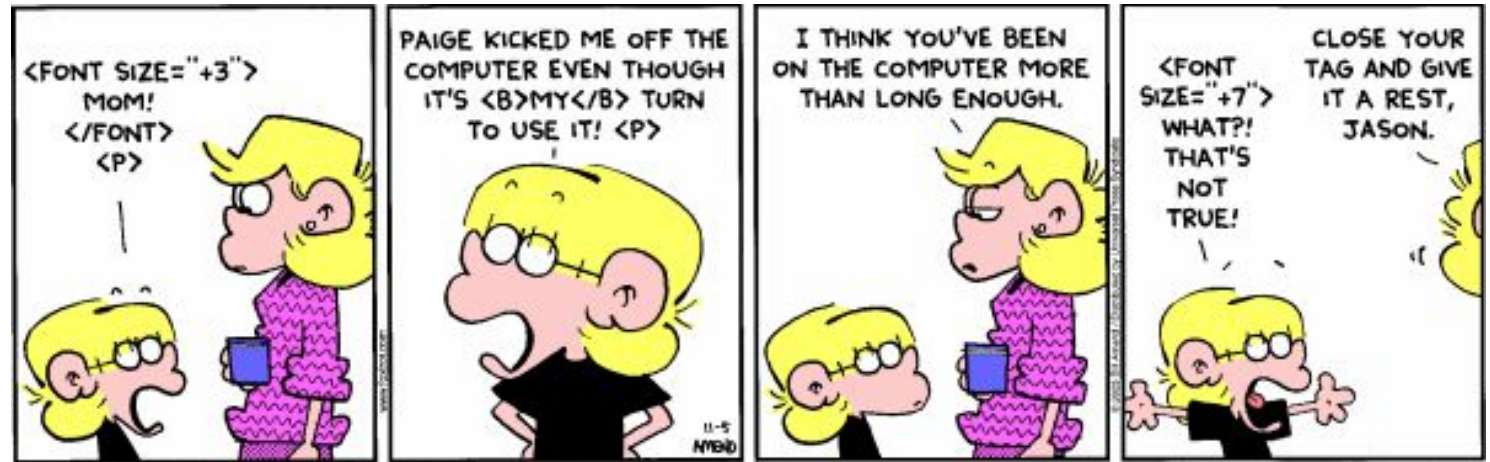


CSE 301

LECTURE 1: BASIC HTML AND CSS



The Internet



- Wikipedia: <http://en.wikipedia.org/wiki/Internet>
- a connection of computer networks using the Internet Protocol (IP)
- layers of communication protocols: IP → TCP/UDP → HTTP/FTP/POP/SMTP/SSH...

Web servers and browsers

- **web server:** software that listens for web page requests
 - [Apache](#)
 - Microsoft Internet Information Server (IIS) ([part of Windows](#))
- **web browser:** fetches/displays documents from web servers
 - [Mozilla Firefox](#)
 - Microsoft [Internet Explorer](#) (IE)
 - Apple [Safari](#)
 - [Google Chrome](#)
 - [Opera](#)

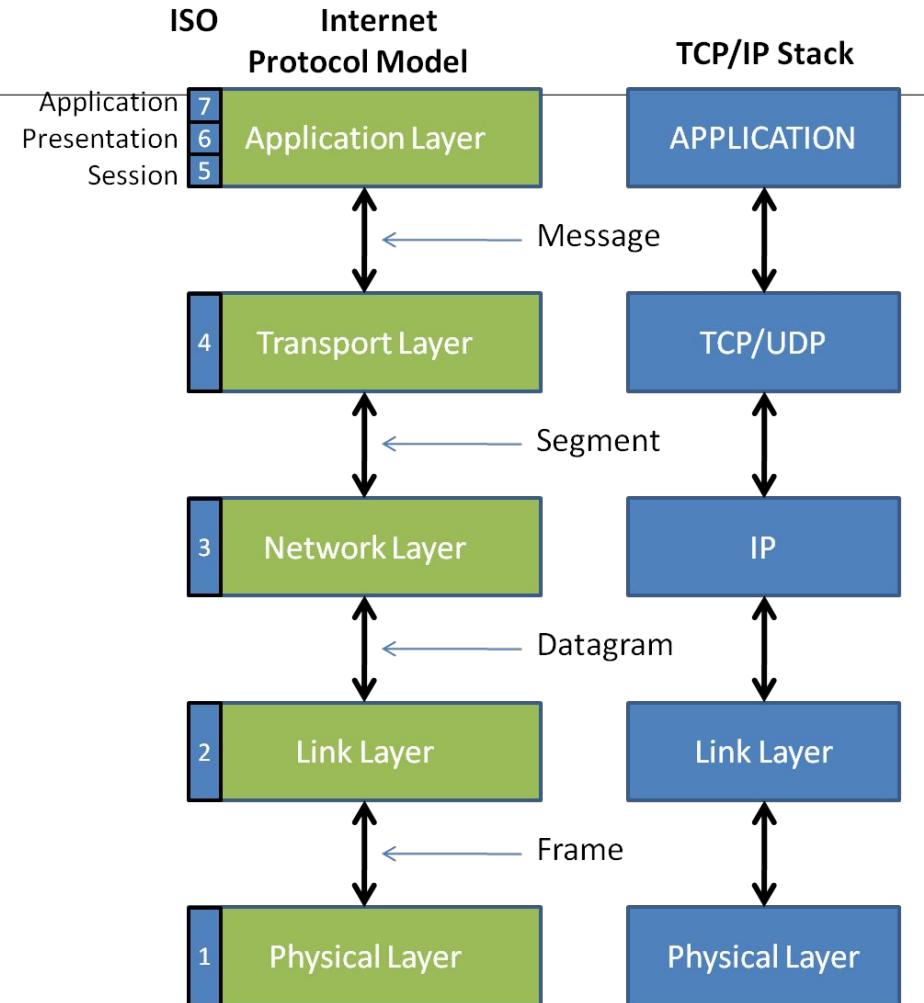
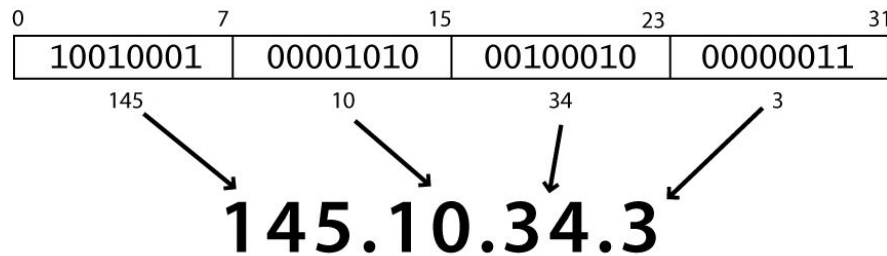


Layers of protocol

Protocol:

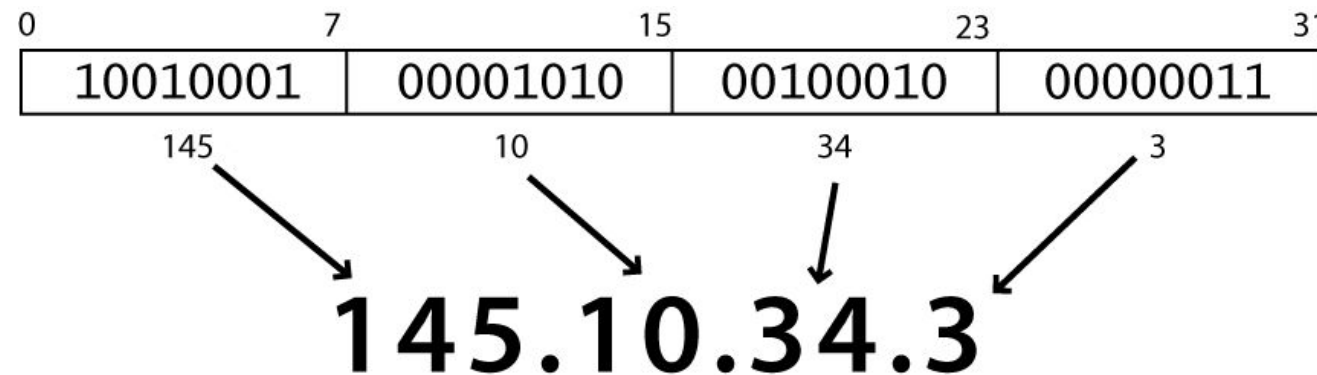
a set of rules governing the format of data sent

IP address:



Internet Protocol (IP)

- a simple protocol for attempting to send data between two computers
- each device has a 32-bit IP address written as four 8-bit numbers (0-255)



- find out your internet IP address: whatismyip.com
- find out your local IP address:
 - in a terminal, type: ipconfig (Windows) or ifconfig (Mac/Linux)

Transmission Control Protocol (TCP)

- adds multiplexing, guaranteed message delivery on top of IP
- **multiplexing**: multiple programs using the same IP address
 - **port**: a number given to each program or service
 - port 80: web browser (port 443 for secure browsing)
 - port 25: email
 - port 22: ssh
 - port 5190: AOL Instant Messenger
 - more common ports
- some programs (games, streaming media programs) use simpler UDP protocol instead of TCP

How do web addresses work?

DNS:

- a set of servers that map written names to IP addresses
 - Example: `www.cs.washington.edu` → `128.208.3.88`
- many systems maintain a local cache called a hosts file

URL:

- a basic URL: `http://www.aw-bc.com/info/regesstepp/index.html`
~~~~ ~~~~~~ ~~~~~~  
protocol host path

# Domain Name System (DNS)

---

- a set of servers that map written names to IP addresses
  - Example: `www.cs.washington.edu` → `128.208.3.88`
- many systems maintain a local cache called a hosts file
  - Windows: `C:\Windows\system32\drivers\etc\hosts`
  - Mac: `/private/etc/hosts`
  - Linux: `/etc/hosts`

# Uniform Resource Locator (URL)

---

- an identifier for the location of a document on a web site
- a basic URL: <http://www.aw-bc.com/info/regesstepp/index.html>

~~~~ ~~~~~~ ~~~~~~  
protocol host path

- upon entering this URL into the browser, it would:
 - ask the DNS server for the IP address of `www.aw-bc.com`
 - connect to that IP address at port 80
 - ask the server to `GET /info/regesstepp/index.html`
 - display the resulting page on the screen

Hypertext Transport Protocol (HTTP)

- the set of commands understood by a web server and sent from a browser
- some HTTP commands (your browser sends these internally):
 - GET **filename** : download
 - POST **filename** : send a web form response
 - PUT **filename** : upload
- simulating a browser with a terminal window:

```
$ telnet www.cs.washington.edu 80
Trying 128.208.3.88...
Connected to 128.208.3.88 (128.208.3.88).
Escape character is '^]'.
GET /index.html
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.0 ...">
<html>
...
```

Who "runs" the internet?

- Internet Engineering Task Force ([IETF](#)): internet protocol standards
- Internet Corporation for Assigned Names and Numbers ([ICANN](#)):
decides top-level [domain names](#)
- World Wide Web Consortium ([W3C](#)): web standards



Brief History

- began as a US Department of Defense network called [ARPANET](#) (1960s-70s)
- initial services: electronic mail, file transfer
- opened to commercial interests in late 80s
- WWW created in 1989-91 by [Tim Berners-Lee](#)
- popular web browsers released: Netscape 1994, IE 1995
- Amazon.com opens in 1995; Google January 1996
- [Hamster Dance](#) web page created in 1999



Web languages / technologies

- Hypertext Markup Language ([HTML](#)): used for writing web pages
- Cascading Style Sheets ([CSS](#)): stylistic info for web pages
- PHP Hypertext Processor ([PHP](#)): dynamically create pages on a web server
- [JavaScript](#): interactive and programmable web pages
- Asynchronous JavaScript and XML ([Ajax](#)): accessing data for web applications
- eXtensible Markup Language ([XML](#)): metalanguage for organizing data
- Structured Query Language ([SQL](#)): interaction with databases

Hypertext Markup Language (HTML)

- describes the *content* and structure of information on a web page
 - not the same as the *presentation* (appearance on screen)
- surrounds text content with opening and closing tags
- each tag's name is called an element
 - syntax: `<element> content </element>`
 - example: `<p>This is a paragraph</p>`
- most whitespace is insignificant in HTML (ignored or collapsed to a single space)
- we will use a newer version called HTML5

Structure of an HTML page

```
<!DOCTYPE html>
<html>
  <head>
    information about the page
  </head>
  <body>
    page contents
  </body>
</html>
```

Page title: `<title>`

describes the title of the web page

`<title>`Chapter 2: HTML Basics`</title>`

- placed within the head of the page
- displayed in the web browser's title bar and when bookmarking the page

Paragraph: <p>

paragraphs of text (block)

```
<p>You're not your job.  
You're not how much money you have in the bank.  
You're not the car you drive.    You're not the contents  
of your wallet. You're not your          khakis.  You're  
the all-singing, all-dancing crap of the world.</p>
```

You're not your job. You're not how much money you have in the bank. You're not the car you drive. You're not the contents of your wallet. You're not your khakis. You're the all-singing, all-dancing crap of the world.

- placed within the body of the page
- [more paragraph examples](#)

Headings: `<h1>`, `<h2>`, ..., `<h6>`

headings to separate major areas of the page (block)

```
<h1>University of Whoville</h1>
```

```
<h2>Department of Computer Science</h2>
```

```
<h3>Sponsored by Micro$oft</h3>
```

University of Whoville

Department of Computer Science

Sponsored by Micro\$oft

- [More heading examples](#)